

CSMProRail2026 — Aegis Rail Protection Proposal

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to:** CSMProRail2026 – Aegis Rail Protection Proposal V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications and measured data invalidate V1.0 specifications
 2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) exceeds ~115 (V1.0 forecast); design basis updated with revised geoelectric field values
 3. **Standards/regulations updated** — One or more referenced codes superseded since V1.0
-

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Rail GIC quantification	General GIC threat described	LA-SF rail (615 km) at LA Basin amplified field: 26,445 V DC, 1,524 A, 40 MW total	Amplified LA Basin field (SC25 adjusted) increases threat 4.3× vs national average assumed in V1.0	USGS 2024; NOAA SWPC Q1 2026
BFRP catenary mast matrix	Thermoset epoxy (no recyclability pathway)	Elium® thermoplastic (95% fiber + 100% monomer recovery via solvolysis 350°C)	Phoenix Protocol mandate; thermoset masts have no viable EOL recovery	Arkema Elium® TDS 2025
Caltrain substation hardening	DC blocking capacitor concept described	500 µF, 12 kV ZrO ₂ ceramic capacitor bank at each Caltrain neutral; \$2.04M total cost	Specific implementation path, component spec, and cost now documented for 11 substations	Caltrain 2×25 kV topology 2025 data

Parameter	V1.0	V2.0	Reason	Primary Source
Signal equipment shielding	Steel cabinet SE ~45 dB	Ti ₃ C ₂ T _x MXene interior spray SE 92 dB (\$460/cabinet)	47 dB deficit eliminates compliance with MIL-STD-461; MXene resolves at confirmed cost	ACS Appl. Mater. Interfaces 2025
Track circuit replacement	Concept referenced	LoRa axle counter mesh: \$8,400/km vs \$12,000/km conventional track circuit, GPS-free	Conventional track circuits use DC through rail — exactly the GIC path being eliminated	LoRa Alliance v1.1
Solar Cycle 25	Generic solar maximum framing	SSN ~137; Caltrain catenary GIC voltage +20% vs V1.0 (26,445 V vs 22,038 V)	Risk quantification requires updated SSN basis for all design voltages	NOAA SWPC Q1 2026

Impact If V1.0 Retained

Parameter	Consequence of Non-Upgrade
Rail GIC quantification	Amplified LA Basin field (SC25 adjusted) increases threat 4.3× vs national average assumed in V1.0
BFRP catenary mast matrix	Phoenix Protocol mandate; thermoset masts have no viable EOL recovery
Caltrain substation hardening	Specific implementation path, component spec, and cost now documented for 11 substations
Signal equipment shielding	47 dB deficit eliminates compliance with MIL-STD-461; MXene resolves at confirmed cost
Track circuit replacement	Conventional track circuits use DC through rail — exactly the GIC path being eliminated
Solar Cycle 25	Risk quantification requires updated SSN basis for all design voltages

Unchanged Elements

All design philosophy, foundational GIC physics equations, product architecture, assembly methodology, and Phoenix Protocol circular economy structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMProRail2026 — Aegis Rail Protection Proposal V1.0 archived; V2.0 is the active specification as of June 2026.